

1. Scenario: You are working on a project that involves analyzing student performance data for a class of 32 students. The data is stored in a NumPy array named `student_scores`, where each row represents a student and each column represents a different subject. The subjects are arranged in the following order: Math, Science, English, and History. Your task is to calculate the average score for each subject and identify the subject with the highest average score.

Question: How would you use NumPy arrays to calculate the average score for each subject and determine the subject with the highest average score? Assume a 4x4 matrix that stores marks of each student in given order.

```
import numpy as np

# Input: marks of 4 students

student_scores = np.array([ [85, 78, 92, 88],[90, 82, 89, 84],[76, 85, 95, 80], [88, 90, 87, 92] ])

# Calculate average for each subject

subject_averages = np.mean(student_scores, axis=0)

subjects = ["Math", "Science", "English", "History"]

# Find subject with highest average

highest_index = np.argmax(subject_averages)

print("Average score for each subject:")

for i in range(4):

    print(subjects[i], ":", subject_averages[i])

print("Subject with highest average:", subjects[highest_index])

===== RESTART: C:/Users/Jannathul firdaus/exp 1 fods.py =====
Average score for each subject:
Math : 84.75
Science : 83.75
English : 90.75
History : 86.0
Subject with highest average: English
>
```

2. Scenario: You are a data analyst working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a NumPy array.

Question: How would you find the average price of all the products sold in the past month?

Assume 3x3 matrix with each row representing the sales for a different product

```
import numpy as np

# Input: sales data

sales_data = np.array([ [100, 150, 200], [250, 300, 350], [400, 450, 500] ])

# Calculate average price of all products

average_price = np.mean(sales_data)

print("Average price of all products:", average_price)

===== RESTART: C:/Users/Jannathul firdaus/exp 2 fods.py =====
Average price of all products: 300.0
```

3. Scenario: You are working on a project that involves analyzing a dataset containing information about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a NumPy array named house_data. Each row of the array represents a house, and the columns contain various features such as the number of bedrooms, square footage, and sale price.

Question: Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?

```
import numpy as np

# Input: house data

# Columns: Bedrooms, Square_Feet, Sale_Price

house_data = np.array([[3, 1500, 250000], [5, 2200, 400000], [4, 1800, 300000], [6, 2800, 550000], [5, 2400, 450000], [3, 1600, 275000]])

# Select houses with more than 4 bedrooms

selected_houses = house_data[house_data[:, 0] > 4]

# Extract sale prices

sale_prices = selected_houses[:, 2]

# Calculate average sale price

average_sale_price = np.mean(sale_prices)
```

```
print("Houses with more than 4 bedrooms:")  
  
print(selected_houses)  
  
print("Average sale price:", average_sale_price)
```

```
===== RESTART: C:/Users/Jannathul firdaus/exp 3 fods.py =====  
Houses with more than 4 bedrooms:  
[[ 5 2200 400000]  
 [ 6 2800 550000]  
 [ 5 2400 450000]]  
Average sale price: 466666.6666666667
```